INTRODUCTION

All living things are made up of many biomolecules such as carbohydrates, proteins, lipids and nucleic acids etc... The major elements present in the human body are carbon, hydrogen, oxygen, nitrogen and phosphorous, and they combine to form a variety of biomolecules. These biomolecules are used as fuel to provide the necessary energy for the various functions of living systems in addition to many other biological functions. The field of studying about the chemistry behind the biological processes is called 'Biochemistry'. In this unit, we will learn about some essential informations of the biomolecules, their structure and their importance.

14.1 Carbohydrates:

Carbohydrates are the most abundant organic compounds in every living organism. They are also known as saccharides (derived from Greek word '*sakcharon*' which means sugar) as many of them are sweet. They are considered as hydrates of carbon, containing hydrogen and oxygen in the same ratio as in water. Chemically, they are defined as polyhydroxy aldehydes or ketones with a general formula $C_n(H_2O)_n$. Some common examples are glucose (monosaccharide), sucrose (disaccharide) and starch (polysaccharide)

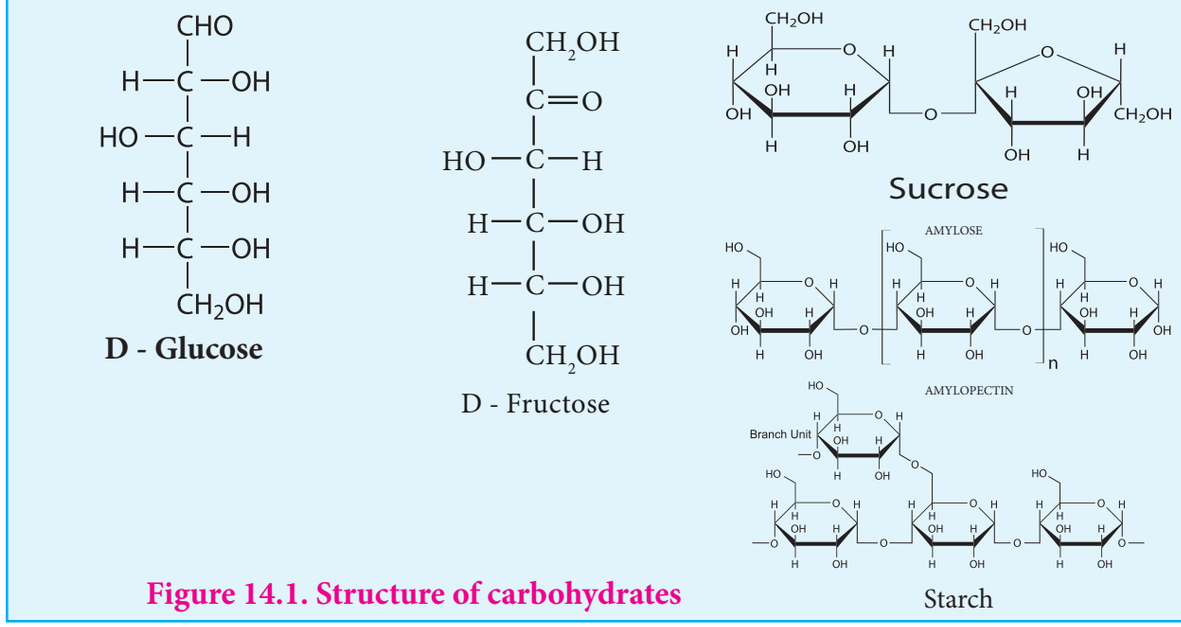


Figure 14.1. Structure of carbohydrates

Carbohydrates are synthesised by green leaves during photo synthesis, a complex process in which sun light provides the energy to convert carbon dioxide and water into glucose and oxygen. Glucose is then converted into other carbohydrates and is consumed by animals.

$$6CO_2 + 6\,H_2O \xrightarrow{\text{Sun light}} C_6H_{12}O_6 + 6O_2$$

14.1.1 Configuration of carbohydrates:

Almost all carbohydrates are optically active as they have one or more chiral carbons. The number of optical isomers depends on the number of chiral carbons (2^n isomers, where n is the total number of chiral carbons). We have already learnt in XI standard to represent an

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